1. Structure of the Human Eye 🔍

Human eye ek natural camera ki tarah kaam karta hai jo objects se aane wali light ko focus karke retina par image banata hai.

Main Parts and Functions:

- **Cornea:** Yeh eye ka transparent outer front part hai. Light iske through enter karti hai aur maximum refraction yahi hoti hai.
- **Iris:** Dark muscular diaphragm jo pupil ke size ko control karta hai. Eye ka colour (jaise blue, black, brown eyes) iris ki wajah se hota hai.
- **Pupil:** Iris ke beech ka chota hole jo enter hone wali light ki amount ko regulate karta hai (Dim light mein bada ho jata hai, bright light mein chota).
- **Crystalline Lens:** Convex lens jo retina par real aur inverted image banata hai.
- **Ciliary Muscles:** Yeh lens ke curvature (focal length) ko adjust karti hain taaki hum pass aur dur ki cheezein saaf dekh sakein.
- **Retina:** Eye ki inner back membrane jahan image banti hai. Isme **Rods** (intensity/brightness ke liye) aur **Cones** (colours ke liye) cells hote hain.
- **Optic Nerve:** Electrical signals ko retina se brain tak bhejti hai.

2. Power of Accommodation (Samayojan Shakti) 📐

Definition: The ability of the eye lens to adjust its focal length so as to clearly see objects at varying distances (near and far) is called **Power of Accommodation**.

- **Near Point (Least distance of distinct vision):** Normal human eye ke liye ye distance **25 cm** hota hai.
- **Far Point:** Normal human eye ke liye infinity (∞) hota hai.

3. Defects of Vision and Their Correction 👓

Jab eye apni power of accommodation kho deti hai, toh vision defects hote hain:

A. Myopia (Short-sightedness / Nikat Drishti Dosh)

- **Meaning:** Person can see nearby objects clearly, but distant objects appear blurred. Image retina ke *pehle* ban jati hai.
- **Causes:** Excessive curvature of eye lens or elongation of eyeball.
- **Correction:** Using a **Concave Lens (Avatal Lens)** of appropriate power.

Defects of Vision (Contd.) 👓

B. Hypermetropia (Long-sightedness / Door Drishti Dosh)

- **Meaning:** Person can see distant objects clearly, but nearby objects appear blurred. Image retina ke *peeche* banti hai.
- **Causes:** Focal length of eye lens is too long or eyeball has become too small.
- **Correction:** Using a **Convex Lens (Uttal Lens)**.

C. Presbyopia (Jara Drishti Dosh)

- **Meaning:** Age badhne ke sath ciliary muscles weak ho jati hain, jisse near point dur ho jata hai (Old age defect).
- **Correction:** Using **Bifocal Lenses** (upper part concave for distance, lower part convex for reading).

D. Cataract (Motiyaabind)

- **Meaning:** Eye lens cloudy/milky ho jata hai age ke sath, jisse vision lose ho jata hai. Surgery se theek hota hai.

4. Refraction of Light through a Glass Prism 🔺

Prism mein do triangular bases aur three rectangular lateral surfaces hoti hain. Light ray jab prism se pass hoti hai, toh woh base ki taraf bend ho jati hai.

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Angle of Deviation ($\delta$): Incident ray aur emergent ray ke beech ka angle. Formula: $\angle A + \angle D = \angle i + \angle e$ Where $\angle A$ = Angle of Prism, $\angle i$ = Angle of Incidence, $\angle e$ = Angle of Emergence, $\angle D$ = Angle of Deviation.
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5. Dispersion of White Light by a Glass Prism 🌈

Definition: Splitting of white light into its component seven colours (VIBGYOR) when passing through a glass prism is called **Dispersion**.

- **VIBGYOR Trick:** Violet, Indigo, Blue, Green, Yellow, Orange, Red.
- **Reason:** Alag-alag colours ki wavelength aur speed glass mein alag hoti hai. **Red** colour sabse kam bend hota hai (longest wavelength) aur **Violet** sabse zyada bend hota hai (shortest wavelength).

6. Atmospheric Refraction 🌅

Earth ke atmosphere ke different layers ka refractive index alag hota hai, jisse light bend ho jati hai.

- **Twinkling of Stars:** Starlight distant hoti hai aur atmosphere ki density change hone ki wajah se apparent position fluctuate karti hai, jisse taare timtimate hain.
- **Advanced Sunrise & Delayed Sunset:** Sun actual sunrise ke 2 minute pehle aur sunset ke 2 minute baad tak dikhta hai atmospheric refraction ki wajah se.

7. Scattering of Light (Tyndall Effect) 💡

Tyndall Effect: Jab light beam colloidal particles par padti hai, toh uska path visible ho jata hai (jaise jungle mein trees ke beech se light aana).

- **Why is the Sky Blue?** Atmosphere ke molecules chote wavelengths (blue/violet) ko zyada scatter karte hain Rayleigh scattering ki wajah se. Human eye blue ke liye zyada sensitive hai.
- **Why are Danger Signals Red?** Red colour ki wavelength sabse longest hoti hai, isliye ye smoke ya fog mein sabse kam scatter hota hai aur dur se dikhta hai.

8. Exam Short Tricks & Mnemonics 💡

🚀 **Topper's Quick Tricks:**

- **Myopia Correction:** M - O - N (Myopia uses cOncave lens).
- **Hypermetropia Correction:** H - O - V (Hypermetropia uses cOnvex lens).
- **Scattering Rule:** Scattering $\propto \frac{1}{\lambda^4}$ (Wavelength jitni kam hogi, scattering utni hi zyada hogi!).

⚡ Quick Revision Checklist (Last-Minute Summary) ✅

- [ ] **Near point:** 25 cm | **Far point:** Infinity.
- [ ] **Myopia:** Concave lens (Nikat drishti dosh).
- [ ] **Hypermetropia:** Convex lens (Door drishti dosh).
- [ ] **Dispersion:** Splitting of light into 7 colours (VIBGYOR).
- [ ] **Red light:** Longest wavelength, minimum scattering.

🎉 **All the Best for Your Board Exams! Score 100/100!** 🌟